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============================================================
 STAT ARB WITH REGIME-SWITCHING + EXECUTION OPTIMISATION
              + PHYSICAL/SOFTWARE ACTION MAPPING
                    [SIMULATION MODE - DEFAULT]
============================================================
[MOCK GPIO] Mode set to BOARD
[MOCK GPIO] Pin 17 set to OUT
[MOCK GPIO] Pin 27 set to OUT
[MOCK GPIO] Pin 22 set to OUT

[INFO] Starting live data stream (simulated ticks)...
[INFO] Each tick prints: prices, regime, PnL, and
physical actions.

[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T001] P1:108.05 P2:90.21 Spread:-0.0550 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T002] P1:125.75 P2:104.75 Spread:-0.0189 Z:0.00
Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T003] P1:117.94 P2:98.28 Spread:-0.0138 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T004] P1:125.14 P2:104.03 Spread:0.2442 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T005] P1:138.27 P2:115.04 Spread:0.2171 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
```

🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T006] P1:146.25 P2:121.48 Spread:0.2975 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T007] P1:118.67 P2:98.55 Spread:0.2852 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T008] P1:113.44 P2:94.28 Spread:0.2309 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T009] P1:107.09 P2:89.10 Spread:0.2286 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T010] P1:109.92 P2:91.41 Spread:0.2642 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T011] P1:103.37 P2:85.89 Spread:0.2068 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T012] P1:102.13 P2:85.00 Spread:0.2085 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON

🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T013] P1:107.37 P2:89.36 Spread:0.1157 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T014] P1:113.29 P2:94.30 Spread:0.0764 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T015] P1:134.80 P2:112.27 Spread:0.1151 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T016] P1:127.73 P2:106.40 Spread:0.0204 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T017] P1:109.64 P2:91.35 Spread:-0.1176 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T018] P1:106.95 P2:88.99 Spread:-0.0547 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T019] P1:94.55 P2:78.98 Spread:-0.1311 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON

🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T020] P1:110.95 P2:92.57 Spread:0.0319 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T021] P1:100.38 P2:83.85 Spread:-0.1050 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL

--- SUMMARY at T=21 ---
  Realized PnL: 0.00 | Unrealized: 0.00 | Total: 0.00
  Trades: 0 | Wins: 0 | Win Rate: 0.0%
  Total Slippage: 0.0000 | Total Fees: 0.00

[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T022] P1:92.21 P2:76.96 Spread:-0.1883 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T023] P1:108.71 P2:90.54 Spread:-0.2137 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T024] P1:90.39 P2:75.51 Spread:-0.1003 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T025] P1:97.46 P2:81.30 Spread:-0.0790 Z:0.00 Regime:0
Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON

🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T026] P1:116.99 P2:97.58 Spread:-0.0054 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T027] P1:125.62 P2:104.60 Spread:0.1820 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T028] P1:151.97 P2:126.56 Spread:0.2317 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T029] P1:169.32 P2:141.06 Spread:0.0651 Z:0.00 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T030] P1:138.43 P2:115.25 Spread:0.1854 Z:0.75 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100% to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T031] P1:135.63 P2:112.97 Spread:0.1618 Z:0.55 Regime:0 Pos:FLAT PnL:0.00 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T032] P1:112.54 P2:93.65 Spread:0.2023 Z:0.77 Regime:1 Pos:FLAT PnL:0.00 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON

🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T033] P1:113.45 P2:94.26 Spread:0.4505 Z:2.13 Regime:1
Pos:SHORT PnL:-0.26 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T034] P1:122.57 P2:102.08 Spread:0.5432 Z:2.40 Regime:1
Pos:SHORT PnL:-0.56 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T035] P1:125.89 P2:104.29 Spread:0.5578 Z:2.21 Regime:1
Pos:SHORT PnL:-0.87 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T036] P1:95.17 P2:78.27 Spread:1.0919 Z:3.61 Regime:1
Pos:SHORT PnL:-1.47 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T037] P1:71.80 P2:58.89 Spread:1.0931 Z:2.95 Regime:1
Pos:SHORT PnL:-2.08 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T038] P1:64.28 P2:52.36 Spread:1.4216 Z:3.15 Regime:1
Pos:SHORT PnL:-2.88 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T039] P1:58.09 P2:46.93 Spread:1.7532 Z:3.15 Regime:1
Pos:SHORT PnL:-3.85 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T040] P1:74.44 P2:60.72 Spread:1.7073 Z:2.59 Regime:1
Pos:SHORT PnL:-4.80 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF

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    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T040] P1:74.44 P2:60.72 Spread:1.7073 Z:2.59 Regime:1
Pos:SHORT PnL:-4.80 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T041] P1:65.89 P2:53.70 Spread:1.4859 Z:1.96 Regime:1
Pos:SHORT PnL:-5.63 Alert: CAUTION
------------------------------------------------------------
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--- SUMMARY at T=41 ---
  Realized PnL: -4.80 | Unrealized: -0.83 | Total: -5.63
  Trades: 8 | Wins: 0 | Win Rate: 0.0%
  Total Slippage: 0.0056 | Total Fees: 0.90
------------------------------------------------------------
------------------------
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T042] P1:53.19 P2:42.85 Spread:1.7751 Z:2.17 Regime:1
Pos:SHORT PnL:-6.63 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T043] P1:56.61 P2:45.42 Spread:2.1165 Z:2.35 Regime:1
Pos:SHORT PnL:-7.82 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T044] P1:47.03 P2:37.17 Spread:2.3437 Z:2.33 Regime:1
Pos:SHORT PnL:-9.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T045] P1:51.41 P2:41.25 Spread:1.8699 Z:1.57 Regime:1
Pos:SHORT PnL:464.63 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T046] P1:53.05 P2:42.86 Spread:1.5987 Z:1.15 Regime:1
Pos:SHORT PnL:735.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
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    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T047] P1:55.57 P2:44.92 Spread:1.6957 Z:1.19 Regime:1
Pos:SHORT PnL:638.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T048] P1:71.60 P2:58.00 Spread:2.0277 Z:1.47 Regime:1
Pos:SHORT PnL:306.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T049] P1:60.63 P2:48.85 Spread:2.0566 Z:1.40 Regime:1
Pos:SHORT PnL:277.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T050] P1:69.93 P2:56.85 Spread:1.7613 Z:0.99 Regime:1
Pos:SHORT PnL:573.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T051] P1:94.77 P2:77.66 Spread:1.5703 Z:0.71 Regime:1
Pos:SHORT PnL:764.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T052] P1:106.75 P2:87.57 Spread:1.7342 Z:0.85 Regime:1
Pos:SHORT PnL:600.33 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T053] P1:137.29 P2:113.07 Spread:1.6766 Z:0.72 Regime:1
Pos:SHORT PnL:657.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T054] P1:141.59 P2:116.90 Spread:1.4341 Z:0.37 Regime:1
Pos:SHORT PnL:900.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
```

```
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T055] P1:146.77 P2:121.04 Spread:1.5846 Z:0.51 Regime:1
Pos:SHORT PnL:749.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T056] P1:142.19 P2:117.60 Spread:1.2782 Z:0.03 Regime:1
Pos:SHORT PnL:1056.33 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T057] P1:133.50 P2:110.39 Spread:0.7793 Z:-0.74
Regime:1 Pos:SHORT PnL:1555.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T058] P1:158.29 P2:130.95 Spread:0.9270 Z:-0.58
Regime:1 Pos:SHORT PnL:1407.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T059] P1:158.64 P2:131.34 Spread:0.9089 Z:-0.69
Regime:1 Pos:SHORT PnL:1425.63 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T060] P1:208.38 P2:172.87 Spread:1.1363 Z:-0.39
Regime:1 Pos:SHORT PnL:1198.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T061] P1:192.63 P2:160.12 Spread:0.5144 Z:-1.56
Regime:1 Pos:SHORT PnL:1820.13 Alert: CAUTION
------------------------------------------------------------
------------------------
--- SUMMARY at T=61 ---
  Realized PnL: -7.82 | Unrealized: 1827.95 | Total:
1820.13
  Trades: 11 | Wins: 0 | Win Rate: 0.0%
  Total Slippage: 0.0092 | Total Fees: 1.20
------------------------------------------------------------
```

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------------------------
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T062] P1:154.04 P2:127.56 Spread:0.9052 Z:-0.96
Regime:1 Pos:SHORT PnL:1429.33 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T063] P1:192.29 P2:159.37 Spread:1.0183 Z:-0.81
Regime:1 Pos:SHORT PnL:1316.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T064] P1:191.95 P2:159.00 Spread:1.0220 Z:-0.88
Regime:1 Pos:SHORT PnL:1312.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T065] P1:231.18 P2:191.77 Spread:0.9116 Z:-1.19
Regime:1 Pos:SHORT PnL:1422.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T066] P1:205.74 P2:170.78 Spread:0.6745 Z:-1.62
Regime:1 Pos:SHORT PnL:1660.03 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T067] P1:196.63 P2:163.85 Spread:0.4136 Z:-2.00
Regime:1 Pos:SHORT PnL:1920.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T068] P1:287.88 P2:239.61 Spread:0.2397 Z:-2.10
Regime:1 Pos:SHORT PnL:2094.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
```

[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T069] P1:221.19 P2:183.55 Spread:0.4851 Z:-1.51
Regime:1 Pos:SHORT PnL:1849.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T070] P1:217.85 P2:180.65 Spread:0.6628 Z:-1.11
Regime:1 Pos:SHORT PnL:1671.73 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T071] P1:245.56 P2:203.79 Spread:0.6061 Z:-1.14
Regime:1 Pos:SHORT PnL:1728.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T072] P1:180.27 P2:149.50 Spread:0.6789 Z:-0.95
Regime:1 Pos:SHORT PnL:1655.63 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T073] P1:153.84 P2:127.91 Spread:0.4961 Z:-1.19
Regime:1 Pos:SHORT PnL:1838.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T074] P1:141.51 P2:117.69 Spread:0.2526 Z:-1.56
Regime:1 Pos:SHORT PnL:2081.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T075] P1:135.86 P2:113.13 Spread:0.0760 Z:-1.74
Regime:1 Pos:SHORT PnL:2258.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T076] P1:131.47 P2:109.60 Spread:0.0419 Z:-1.66
Regime:1 Pos:SHORT PnL:2292.63 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON

🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T077] P1:161.43 P2:134.78 Spread:-0.3403 Z:-2.09
Regime:1 Pos:SHORT PnL:2674.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T078] P1:173.30 P2:144.63 Spread:-0.0963 Z:-1.59
Regime:1 Pos:SHORT PnL:2430.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T079] P1:142.83 P2:119.07 Spread:-0.0237 Z:-1.41
Regime:1 Pos:SHORT PnL:2358.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T080] P1:178.39 P2:148.55 Spread:-0.0238 Z:-1.34
Regime:1 Pos:SHORT PnL:2358.33 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T081] P1:209.43 P2:174.91 Spread:-0.5237 Z:-2.04
Regime:1 Pos:SHORT PnL:2858.23 Alert: CAUTION

--- SUMMARY at T=81 ---
  Realized PnL: -7.82 | Unrealized: 2866.05 | Total:
2858.23
  Trades: 11 | Wins: 0 | Win Rate: 0.0%
  Total Slippage: 0.0092 | Total Fees: 1.20

[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T082] P1:203.48 P2:169.48 Spread:-0.3577 Z:-1.66
Regime:1 Pos:SHORT PnL:2692.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T083] P1:179.69 P2:149.99 Spread:-0.3170 Z:-1.51
Regime:1 Pos:SHORT PnL:2651.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF

```
[T083] P1:179.69 P2:149.99 Spread:-0.3170 Z:-1.51
Regime:1 Pos:SHORT PnL:2651.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T084] P1:187.27 P2:156.17 Spread:-0.2323 Z:-1.28
Regime:1 Pos:SHORT PnL:2566.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T085] P1:277.37 P2:231.80 Spread:-0.6502 Z:-1.95
Regime:1 Pos:SHORT PnL:2984.73 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T086] P1:242.26 P2:201.99 Spread:-0.5333 Z:-1.62
Regime:1 Pos:SHORT PnL:2867.83 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T087] P1:291.94 P2:243.58 Spread:-0.9230 Z:-2.10
Regime:1 Pos:SHORT PnL:3257.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T088] P1:327.31 P2:273.80 Spread:-0.9997 Z:-2.01
Regime:1 Pos:SHORT PnL:3334.23 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T089] P1:317.58 P2:265.14 Spread:-0.7754 Z:-1.52
Regime:1 Pos:SHORT PnL:3109.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T090] P1:334.96 P2:279.89 Spread:-0.6675 Z:-1.27
Regime:1 Pos:SHORT PnL:3002.03 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
```

🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T091] P1:391.33 P2:326.63 Spread:-0.4782 Z:-0.89
Regime:1 Pos:SHORT PnL:2812.73 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T092] P1:368.74 P2:307.53 Spread:0.0031 Z:-0.03
Regime:1 Pos:SHORT PnL:2331.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T093] P1:356.93 P2:296.71 Spread:0.0846 Z:0.17 Regime:1
Pos:SHORT PnL:2249.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T094] P1:318.34 P2:265.90 Spread:-0.3890 Z:-0.65
Regime:1 Pos:SHORT PnL:2723.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T095] P1:330.51 P2:275.45 Spread:-0.3124 Z:-0.44
Regime:1 Pos:SHORT PnL:2646.93 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T096] P1:351.16 P2:292.80 Spread:0.0900 Z:0.45 Regime:1
Pos:SHORT PnL:2244.53 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T097] P1:305.36 P2:253.98 Spread:0.6471 Z:1.60 Regime:1
Pos:SHORT PnL:1687.43 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T098] P1:374.34 P2:311.85 Spread:0.8127 Z:1.81 Regime:1
Pos:SHORT PnL:1686.98 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF

[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T099] P1:393.24 P2:326.62 Spread:0.7197 Z:1.58 Regime:1
Pos:SHORT PnL:1779.98 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T100] P1:341.80 P2:284.16 Spread:0.8791 Z:1.83 Regime:1
Pos:SHORT PnL:1779.47 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T101] P1:366.96 P2:305.70 Spread:0.6112 Z:1.32 Regime:1
Pos:SHORT PnL:2047.37 Alert: CAUTION

--- SUMMARY at T=101 ---
  Realized PnL: 1779.98 | Unrealized: 267.40 | Total:
2047.37
  Trades: 13 | Wins: 2 | Win Rate: 15.4%
  Total Slippage: 0.0101 | Total Fees: 1.40

[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T102] P1:342.07 P2:284.54 Spread:0.4800 Z:1.10 Regime:1
Pos:SHORT PnL:2178.57 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T103] P1:430.41 P2:358.51 Spread:0.0184 Z:0.23 Regime:1
Pos:SHORT PnL:2640.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T104] P1:373.50 P2:310.41 Spread:-0.0355 Z:0.14
Regime:1 Pos:SHORT PnL:2694.07 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T105] P1:391.57 P2:326.47 Spread:-0.0804 Z:0.06
Regime:1 Pos:SHORT PnL:2738.97 Alert: CAUTION

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[T105] P1:391.57 P2:326.47 Spread:-0.0804 Z:0.06
Regime:1 Pos:SHORT PnL:2738.97 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T106] P1:423.25 P2:352.92 Spread:-0.3283 Z:-0.41
Regime:1 Pos:SHORT PnL:2986.87 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T107] P1:414.31 P2:345.28 Spread:-0.2525 Z:-0.26
Regime:1 Pos:SHORT PnL:2911.07 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T108] P1:544.36 P2:455.05 Spread:-0.1736 Z:-0.10
Regime:1 Pos:SHORT PnL:2832.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T109] P1:524.10 P2:437.06 Spread:-0.0872 Z:0.08
Regime:1 Pos:SHORT PnL:2745.77 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T110] P1:634.43 P2:527.97 Spread:0.1537 Z:0.55 Regime:1
Pos:SHORT PnL:2504.87 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T111] P1:678.01 P2:564.77 Spread:0.5705 Z:1.29 Regime:1
Pos:SHORT PnL:2088.07 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T112] P1:609.16 P2:507.38 Spread:0.5826 Z:1.23 Regime:1
Pos:SHORT PnL:2075.97 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
```

```
[T113] P1:539.11 P2:448.29 Spread:0.2256 Z:0.50 Regime:1
Pos:SHORT PnL:2432.97 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T114] P1:650.65 P2:542.42 Spread:0.0508 Z:0.15 Regime:1
Pos:SHORT PnL:2607.77 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T115] P1:639.10 P2:531.34 Spread:-0.2869 Z:-0.54
Regime:1 Pos:SHORT PnL:2945.47 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T116] P1:533.97 P2:444.96 Spread:-0.1096 Z:-0.22
Regime:1 Pos:SHORT PnL:2768.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T117] P1:461.22 P2:383.74 Spread:-0.0519 Z:-0.18
Regime:1 Pos:SHORT PnL:2710.47 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T118] P1:418.74 P2:348.76 Spread:-0.1538 Z:-0.50
Regime:1 Pos:SHORT PnL:2812.37 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T119] P1:435.33 P2:362.47 Spread:0.1551 Z:0.17 Regime:1
Pos:SHORT PnL:2503.47 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T120] P1:521.94 P2:435.07 Spread:0.0864 Z:-0.07
Regime:1 Pos:SHORT PnL:2572.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
```

[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T121] P1:512.48 P2:427.43 Spread:-0.3592 Z:-1.30
Regime:1 Pos:SHORT PnL:3017.77 Alert: CAUTION

--- SUMMARY at T=121 ---
Realized PnL: 1779.98 | Unrealized: 1237.80 | Total:
3017.77
Trades: 13 | Wins: 2 | Win Rate: 15.4%
Total Slippage: 0.0101 | Total Fees: 1.40

[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T122] P1:438.01 P2:365.08 Spread:-0.2368 Z:-0.93
Regime:1 Pos:SHORT PnL:2895.37 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T123] P1:618.92 P2:515.84 Spread:-0.4540 Z:-1.41
Regime:1 Pos:SHORT PnL:3112.57 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T124] P1:630.32 P2:525.72 Spread:-0.1516 Z:-0.67
Regime:1 Pos:SHORT PnL:2810.17 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T125] P1:735.70 P2:612.68 Spread:0.1993 Z:0.22 Regime:1
Pos:SHORT PnL:2459.27 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T126] P1:607.66 P2:506.70 Spread:0.2512 Z:0.35 Regime:1
Pos:SHORT PnL:2407.37 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
[ACTION] Regime CAUTION -> Reducing SPEED to 50%.
[SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T127] P1:404.65 P2:337.17 Spread:0.5703 Z:1.22 Regime:1
Pos:SHORT PnL:2088.27 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF

```
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T128] P1:376.73 P2:313.18 Spread:0.5802 Z:1.31 Regime:1
Pos:SHORT PnL:2078.37 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 27 turned ON
    🟡 [ACTION] Regime CAUTION -> Reducing SPEED to 50%.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 50
[T129] P1:405.29 P2:337.45 Spread:1.0444 Z:2.43 Regime:1
Pos:SHORT PnL:2077.79 Alert: CAUTION
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T130] P1:475.57 P2:396.12 Spread:0.8960 Z:2.03 Regime:0
Pos:SHORT PnL:2077.44 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T131] P1:365.34 P2:304.20 Spread:0.5934 Z:1.24 Regime:0
Pos:SHORT PnL:2380.04 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T132] P1:325.63 P2:271.08 Spread:0.4040 Z:0.75 Regime:0
Pos:SHORT PnL:2569.44 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T133] P1:258.78 P2:215.36 Spread:0.3997 Z:0.70 Regime:0
Pos:SHORT PnL:2573.74 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T134] P1:261.69 P2:217.80 Spread:0.1827 Z:0.11 Regime:0
Pos:SHORT PnL:2790.74 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
```

```
[T134] P1:261.69 P2:217.80 Spread:0.1827 Z:0.11 Regime:0
Pos:SHORT PnL:2790.74 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T135] P1:313.06 P2:260.32 Spread:0.2068 Z:0.15 Regime:0
Pos:SHORT PnL:2766.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T136] P1:503.51 P2:419.32 Spread:-0.0073 Z:-0.46
Regime:0 Pos:SHORT PnL:2980.74 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T137] P1:516.16 P2:430.46 Spread:-0.0902 Z:-0.71
Regime:0 Pos:SHORT PnL:3063.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T138] P1:542.77 P2:453.26 Spread:-0.0481 Z:-0.61
Regime:0 Pos:SHORT PnL:3021.54 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T139] P1:493.80 P2:411.55 Spread:-0.0363 Z:-0.58
Regime:0 Pos:SHORT PnL:3009.74 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T140] P1:529.33 P2:441.41 Spread:-0.1948 Z:-0.98
Regime:0 Pos:SHORT PnL:3168.24 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
```

```
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T141] P1:571.24 P2:476.66 Spread:-0.0835 Z:-0.62
Regime:0 Pos:SHORT PnL:3056.94 Alert: OPTIMAL
------------------------------------------------------------
------------------------
--- SUMMARY at T=141 ---
  Realized PnL: 2077.79 | Unrealized: 979.16 | Total:
3056.94
  Trades: 15 | Wins: 4 | Win Rate: 26.7%
  Total Slippage: 0.0111 | Total Fees: 1.60
------------------------------------------------------------
------------------------
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T142] P1:563.23 P2:469.26 Spread:0.0330 Z:-0.25
Regime:0 Pos:SHORT PnL:2940.44 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T143] P1:450.51 P2:375.37 Spread:0.1098 Z:-0.02
Regime:0 Pos:SHORT PnL:2863.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T144] P1:465.61 P2:388.57 Spread:0.0626 Z:-0.16
Regime:0 Pos:SHORT PnL:2910.84 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T145] P1:495.09 P2:412.35 Spread:0.0708 Z:-0.17
Regime:0 Pos:SHORT PnL:2902.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
   📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T146] P1:492.06 P2:409.75 Spread:-0.0094 Z:-0.42
Regime:0 Pos:SHORT PnL:2982.84 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
   🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
```

```
[T147] P1:582.72 P2:486.00 Spread:-0.0529 Z:-0.55
Regime:0 Pos:SHORT PnL:3026.34 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T148] P1:567.45 P2:473.48 Spread:-0.1680 Z:-0.88
Regime:0 Pos:SHORT PnL:3141.44 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T149] P1:538.11 P2:448.33 Spread:-0.1912 Z:-0.91
Regime:0 Pos:SHORT PnL:3164.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T150] P1:668.29 P2:557.77 Spread:-0.2877 Z:-1.13
Regime:0 Pos:SHORT PnL:3261.14 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T151] P1:460.64 P2:383.84 Spread:-0.0819 Z:-0.58
Regime:0 Pos:SHORT PnL:3055.34 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T152] P1:457.26 P2:380.91 Spread:-0.1727 Z:-0.86
Regime:0 Pos:SHORT PnL:3146.14 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T153] P1:530.98 P2:442.65 Spread:-0.2102 Z:-1.02
Regime:0 Pos:SHORT PnL:3183.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
```

[T154] P1:593.56 P2:495.10 Spread:-0.0572 Z:-0.57
Regime:0 Pos:SHORT PnL:3030.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T155] P1:498.94 P2:416.19 Spread:-0.0490 Z:-0.52
Regime:0 Pos:SHORT PnL:3022.44 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T156] P1:501.82 P2:418.25 Spread:-0.1292 Z:-0.72
Regime:0 Pos:SHORT PnL:3102.64 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T157] P1:545.23 P2:454.25 Spread:-0.0179 Z:-0.34
Regime:0 Pos:SHORT PnL:2991.34 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T158] P1:671.01 P2:559.53 Spread:0.1549 Z:0.26 Regime:0
Pos:SHORT PnL:2818.54 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T159] P1:773.87 P2:645.62 Spread:0.1431 Z:0.39 Regime:0
Pos:SHORT PnL:2830.34 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T160] P1:722.31 P2:602.33 Spread:0.2902 Z:1.33 Regime:0
Pos:SHORT PnL:2683.24 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
🟢 [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100

```
[T161] P1:662.82 P2:553.29 Spread:0.2990 Z:1.60 Regime:0
Pos:SHORT PnL:2674.44 Alert: OPTIMAL
------------------------------------------------------------
------------------------
--- SUMMARY at T=161 ---
  Realized PnL: 2077.79 | Unrealized: 596.66 | Total:
2674.44
  Trades: 15 | Wins: 4 | Win Rate: 26.7%
  Total Slippage: 0.0111 | Total Fees: 1.60
------------------------------------------------------------
------------------------
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T162] P1:721.44 P2:601.21 Spread:0.1414 Z:0.82 Regime:0
Pos:SHORT PnL:2832.04 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T163] P1:808.33 P2:674.45 Spread:0.2586 Z:1.66 Regime:0
Pos:SHORT PnL:2714.84 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T164] P1:754.63 P2:628.51 Spread:0.3882 Z:2.28 Regime:0
Pos:SHORT PnL:2714.69 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 17 turned ON
    [ACTION] Regime PREFERRED -> Sending SPEED = 100%
to Motor Controller.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 100
[T165] P1:598.25 P2:498.31 Spread:0.2022 Z:1.16 Regime:0
Pos:SHORT PnL:2900.69 Alert: OPTIMAL
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T166] P1:541.19 P2:451.59 Spread:-0.0770 Z:-0.50
Regime:2 Pos:SHORT PnL:3179.89 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
```

[T167] P1:709.77 P2:591.06 Spread:0.6957 Z:3.21 Regime:2
Pos:SHORT PnL:3179.31 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T168] P1:894.34 P2:743.59 Spread:1.5402 Z:4.28 Regime:2
Pos:SHORT PnL:3178.05 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T169] P1:667.65 P2:555.60 Spread:1.2137 Z:2.75 Regime:2
Pos:SHORT PnL:3177.05 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T170] P1:722.24 P2:601.96 Spread:0.7703 Z:1.50 Regime:2
Pos:SHORT PnL:3620.45 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T171] P1:705.34 P2:588.05 Spread:1.3022 Z:2.43 Regime:2
Pos:SHORT PnL:3619.39 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T172] P1:604.22 P2:502.73 Spread:1.2107 Z:1.99 Regime:2
Pos:SHORT PnL:3618.39 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T173] P1:418.29 P2:347.96 Spread:0.9461 Z:1.35 Regime:2
Pos:SHORT PnL:3882.99 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5

[T174] P1:358.05 P2:296.83 Spread:1.4812 Z:2.14 Regime:2
Pos:SHORT PnL:3881.74 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T175] P1:327.70 P2:272.06 Spread:1.2224 Z:1.53 Regime:2
Pos:SHORT PnL:4140.54 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T176] P1:331.20 P2:275.31 Spread:0.9735 Z:1.02 Regime:2
Pos:SHORT PnL:4389.44 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T177] P1:384.44 P2:319.66 Spread:1.2070 Z:1.33 Regime:2
Pos:SHORT PnL:4155.94 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T178] P1:396.71 P2:329.64 Spread:1.0156 Z:0.93 Regime:2
Pos:SHORT PnL:4347.34 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T179] P1:432.34 P2:359.24 Spread:1.5187 Z:1.66 Regime:2
Pos:SHORT PnL:3844.24 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T180] P1:338.03 P2:281.42 Spread:0.8493 Z:0.49 Regime:2
Pos:SHORT PnL:4513.64 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5

```
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T181] P1:336.45 P2:279.89 Spread:0.6808 Z:0.15 Regime:2
Pos:SHORT PnL:4682.14 Alert: DANGER / STOP
------------------------------------------------------------
------------------------
--- SUMMARY at T=181 ---
  Realized PnL: 3882.99 | Unrealized: 799.14 | Total:
4682.14
  Trades: 22 | Wins: 11 | Win Rate: 50.0%
  Total Slippage: 0.0174 | Total Fees: 2.30
------------------------------------------------------------
------------------------
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T182] P1:320.61 P2:266.82 Spread:1.1581 Z:0.93 Regime:2
Pos:SHORT PnL:4204.84 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T183] P1:251.03 P2:207.85 Spread:1.0600 Z:0.71 Regime:2
Pos:SHORT PnL:4302.94 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T184] P1:266.37 P2:221.34 Spread:0.9124 Z:0.38 Regime:2
Pos:SHORT PnL:4450.54 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T185] P1:277.85 P2:230.12 Spread:1.9188 Z:2.09 Regime:2
Pos:SHORT PnL:4448.95 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
    [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T186] P1:275.24 P2:227.97 Spread:2.0385 Z:2.11 Regime:2
Pos:SHORT PnL:4447.29 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
    [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
```

📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T187] P1:284.71 P2:234.61 Spread:2.9210 Z:3.02 Regime:2
Pos:SHORT PnL:4444.88 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T188] P1:202.44 P2:166.49 Spread:3.1011 Z:2.77 Regime:2
Pos:SHORT PnL:4442.31 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T189] P1:263.32 P2:217.34 Spread:2.3378 Z:1.61 Regime:2
Pos:SHORT PnL:5205.61 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T190] P1:301.18 P2:249.07 Spread:2.5088 Z:1.68 Regime:2
Pos:SHORT PnL:5034.61 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T191] P1:269.88 P2:222.74 Spread:2.7208 Z:1.79 Regime:2
Pos:SHORT PnL:4822.61 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T192] P1:245.19 P2:202.55 Spread:2.3162 Z:1.21 Regime:2
Pos:SHORT PnL:5227.21 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T193] P1:194.44 P2:160.16 Spread:2.6034 Z:1.46 Regime:2
Pos:SHORT PnL:4940.01 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced to 5%. Webhook alert triggered.

📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T194] P1:179.56 P2:147.46 Spread:2.7027 Z:1.48 Regime:2
Pos:SHORT PnL:4840.71 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T195] P1:227.72 P2:188.01 Spread:1.8329 Z:0.36 Regime:2
Pos:SHORT PnL:5710.51 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T196] P1:234.97 P2:194.69 Spread:1.5357 Z:-0.10
Regime:2 Pos:SHORT PnL:6007.71 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T197] P1:195.37 P2:162.36 Spread:0.9187 Z:-0.98
Regime:2 Pos:SHORT PnL:6624.71 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T198] P1:200.26 P2:166.54 Spread:0.4879 Z:-1.48
Regime:2 Pos:SHORT PnL:7055.51 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T199] P1:174.96 P2:145.63 Spread:0.4225 Z:-1.48
Regime:2 Pos:SHORT PnL:7120.91 Alert: DANGER / STOP
[MOCK GPIO] >>> Pin 17 turned OFF
[MOCK GPIO] >>> Pin 27 turned OFF
[MOCK GPIO] >>> Pin 22 turned OFF
[MOCK GPIO] >>> Pin 22 turned ON
🔴 [ACTION] Regime WORST -> EMERGENCY! Speed reduced
to 5%. Webhook alert triggered.
📡 [SERIAL] Writing to COM3/Dev/ttyUSB0: SPEED 5
[T200] P1:167.79 P2:139.97 Spread:-0.0426 Z:-1.95
Regime:2 Pos:SHORT PnL:7586.01 Alert: DANGER / STOP

==========
FINAL PERFORMANCE REPORT & REGIME RANKING (PREFERRED →
WORST)
==========

```
============================================================
 FINAL PERFORMANCE REPORT & REGIME RANKING (PREFERRED → WORST)
============================================================

OVERALL SYSTEM PERFORMANCE:
  ▶ Final Total PnL: 7586.01
  ▶ Average Rolling Sharpe: 0.261
  ▶ Maximum Drawdown: -1647.25
  ▶ Total Trades Executed: 26
  ▶ Win Rate: 57.7%
  ▶ Total Execution Slippage: 0.0256
  ▶ Total Fees Paid: 2.70

------------------------------------------------------------
 REGIME RANKING (from PREFERRED / BEST to WORST):
------------------------------------------------------------
 Regime                                Label  Avg Sharpe  Avg Volatility  Avg |Z-Score|  Total PnL  Win Rate  Preference Score
      2    Worst (High Vol, Slow Reversion)       3.040            0.50          1.196    7586.01     0.577             2.887
      1              Neutral (Moderate Vol)       0.506            0.25          0.063    2077.79     0.577             0.593
      0 Preferred (Low Vol, Fast Reversion)      -1.550            0.10          0.041    2900.69     0.577            -0.225

------------------------------------------------------------
 DETAILED BREAKDOWN PER REGIME:
------------------------------------------------------------

▶ Regime 2: Worst (High Vol, Slow Reversion)
   - Avg Sharpe: 3.04
   - Avg Volatility: 0.5
   - Avg |Z-Score|: 1.196
   - Total PnL generated in this regime: 7586.01
   - Win Rate: 57.7%
   - Preference Score: 2.887

▶ Regime 1: Neutral (Moderate Vol)
   - Avg Sharpe: 0.506
   - Avg Volatility: 0.25
   - Avg |Z-Score|: 0.063
   - Total PnL generated in this regime: 2077.79
   - Win Rate: 57.7%
   - Preference Score: 0.593

▶ Regime 0: Preferred (Low Vol, Fast Reversion)
   - Avg Sharpe: -1.55
   - Avg Volatility: 0.1
   - Avg |Z-Score|: 0.041
   - Total PnL generated in this regime: 2900.69
   - Win Rate: 57.7%
   - Preference Score: -0.225

[INFO] Analysis complete. All physical actions were simulated in software.
```